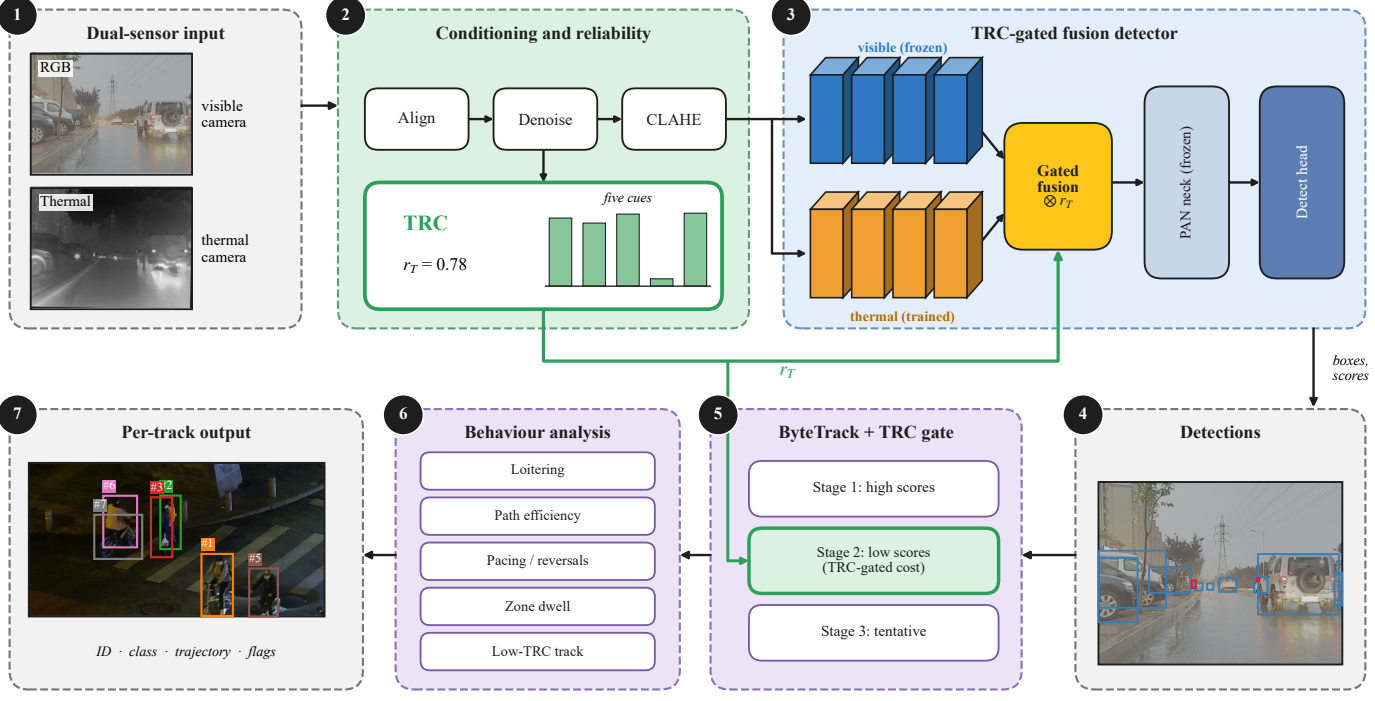

1
Dual-sensor input
RGB
visible camera
Thermal
thermal camera
2
Conditioning and reliability
Align
Denoise
CLAHE
TRC
$r_T = 0.78$
five cues
3
TRC-gated fusion detector
visible (frozen)
thermal (trained)
Gated fusion ⊗ r_T
PAN neck (frozen)
Detect head
r_T
boxes, scores
4
Detections
5
ByteTrack + TRC gate
Stage 1: high scores
Stage 2: low scores (TRC-gated cost)
Stage 3: tentative
6
Behaviour analysis
Loitering
Path efficiency
Pacing / reversals
Zone dwell
Low-TRC track
7
Per-track output
ID · class · trajectory · flags